Mathematical Ways of Measuring Risk

A Survey in Quantitative Finance and Portfolio Theory

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1 Definition of Risk

1.1 Conceptual Foundation: Knight's Distinction

The formal study of risk begins with Frank Knight's seminal 1921 distinction between two concepts that are often conflated:

- **Risk:** Outcomes whose probability distribution is known. The set of possible outcomes and their likelihoods can be fully characterized.
- **Uncertainty:** Outcomes whose probability distribution is itself unknown. No probability can be meaningfully assigned to outcomes.

What financial mathematics calls "risk measurement" operates entirely within Knight's risk category: we assume a probability distribution exists and seek to summarize its characteristics - particularly its downside - in a single number.

1.2 Risk as a Mathematical Functional

Definition 1.1 (Risk Measure)**.** A **risk measure** is a functional ρ that maps a random variable X (representing future profit and loss, or loss) to a single real number $\rho(X)$, interpreted as:

$$\rho(X) \;=\; \text{the capital or buffer required to make position } X \text{ financially acceptable.}$$

Markowitz (1952) was the first to operationalize this idea, equating risk with the dispersion of portfolio returns around their expected value:

$$\text{Risk} = \text{Var}(R) = E\left[(R - E[R])^2\right].$$

This definition was so productive - enabling portfolio theory, CAPM, and the entire modern finance edifice - that it went largely unquestioned for decades.

1.3 The Four Coherence Axioms

Artzner, Delbaen, Eber, and Heath (1999) shifted the question from "which formula should we use for risk?" to "what properties should any reasonable risk measure satisfy?" They proposed four axioms, calling measures that satisfy all four **coherent**.

Let ρ be a risk measure and X, Y be random loss variables, $\lambda > 0$, $c \in \mathbb{R}$.

1. **Monotonicity.** If X always loses at least as much as Y (i.e. $X \geq Y$ a.s.), then $\rho(X) \geq \rho(Y)$. A position with universally worse outcomes must require at least as

much capital.

2. **Subadditivity.**

$$\rho(X+Y) \;\leq\; \rho(X)+\rho(Y).$$

Combining two positions cannot increase total risk beyond the sum of their individual risks. This is the mathematical formalization of "diversification does not increase risk."

3. **Positive Homogeneity.**

$$\rho(\lambda X) \;=\; \lambda\,\rho(X) \quad \text{for all } \lambda > 0.$$

Scaling a position by a factor scales its required capital by the same factor.

4. **Translation Invariance.**

$$\rho(X+c) \;=\; \rho(X) - c.$$

Adding a riskless cash amount c reduces required capital one-for-one.

These four axioms serve as the benchmark against which every risk measure in this document will be evaluated (see Section 7).

2 Dispersion / Moment-Based Measures

2.1 Variance and Standard Deviation

Definition 2.1 (Portfolio Variance)**.**

$$\sigma^2 \;=\; E\Big[\big(R - E[R]\big)^2\Big]$$

Risk is defined as the spread of the return distribution around its mean, treating upside and downside deviations symmetrically.

Markowitz (1952) states explicitly:

> "*We next consider the rule that the investor does (or should) consider expected return a desirable thing and variance of return an undesirable thing.*"

This sentence is his definition of risk - variance equals risk - stated as an assumption and built upon without further justification.

For a portfolio of n assets with weights $x = (x_1, \ldots, x_n)$, $\sum_{i=1}^{n} x_i = 1$ (where x_i is the

fraction of wealth in asset i), portfolio variance is:

$$\sigma_p^2 \;=\; \sum_{i=1}^{n}\sum_{j=1}^{n} x_i\, x_j\, \sigma_{ij}, \qquad \sigma_{ij} = \mathrm{Cov}(R_i, R_j).$$

Reference: Markowitz, H. (1952). *Portfolio Selection.* The Journal of Finance, 7(1), 77–91. **Read:** The section deriving the E-V rule (expected return–variance rule) and the geometric argument for diversification.

Consequences: Variance leads to a quadratic program - tractable and with a clean, closed-form covariance matrix. However, it penalizes upside volatility identically to downside volatility. A portfolio that occasionally returns +30% contributes the same to variance as one that occasionally returns −30%, which does not reflect how investors perceive risk. Variance also implicitly assumes that returns are fully characterized by their first two moments - an assumption exact only under normality.

2.2 Semi-Variance / Downside Deviation

Markowitz himself acknowledged the symmetry flaw of his 1952 model. In his 1959 book he proposed **semi-variance** as a theoretically superior alternative.
Definition 2.2 (Semi-Variance)**.**

$$SV \;=\; E\Big[\min\big(R - E[R],\; 0\big)^2\Big]$$

Only deviations below the mean are squared and averaged. Upside deviations contribute zero.

In 1952, Markowitz penalized upside deviations as much as downside, which is conceptually wrong. Semi-variance corrects this by restricting attention to the outcomes investors actually dislike.

Reference: Markowitz, H. (1959). *Portfolio Selection: Efficient Diversification of Investments.* Wiley. **Read:** The chapter introducing semi-variance.

Consequences: Semi-variance correctly isolates downside risk from upside opportunity. However, it destroys the clean quadratic-programming structure that made variance tractable. With variance, the covariance matrix Σ is constant - independent of which portfolio is being evaluated. With semi-variance, the equivalent co-semi-variance matrix is no longer constant: it changes as the portfolio weights change, because the set of below-target observations itself shifts as weights are altered. This makes optimization iterative and computationally expensive. Markowitz concluded that semi-variance is conceptually superior but practically intractable, and retained

variance as the working measure.

2.3 Mean Absolute Deviation (MAD)

Definition 2.3 (Mean Absolute Deviation - the L^1 Risk Model)**.**

$$W \;=\; E\big[\big|R - E[R]\big|\big]$$

The average of absolute (not squared) deviations from the mean.

Markowitz's variance minimization is a **quadratic program** (QP), which was computationally slow and unstable for large portfolios in 1991. Konno and Yamazaki's entire contribution was to replace variance with MAD, converting the problem into a **linear program** (LP).

Key theoretical result. Under multivariate normal returns, minimizing MAD and minimizing variance yield the exact same efficient frontier. This follows from the identity:

$$E[|R - \mu|] \;=\; \sqrt{\tfrac{2}{\pi}}\,\sigma.$$

Since $\sqrt{2/\pi} \approx 0.798$ is a constant, minimizing $E|R - \mu|$ is equivalent to minimizing σ^2 under normality.

The LP Reformulation. Portfolio variance with weights requires cross-products $x_i x_j$ - making it quadratic. MAD avoids this. Using T historical return scenarios:

$$W \;=\; \frac{1}{T}\sum_{t=1}^{T}\left|\sum_{i=1}^{n} x_i r_{it} - \mu_p\right|, \qquad \mu_p = \sum_{i=1}^{n}\mu_i x_i.$$

The absolute value is non-linear, but introducing auxiliary variables $y_t \geq 0$:

$$\begin{aligned} y_t &\geq \sum_{i=1}^{n} x_i r_{it} - \mu_p, \\ y_t &\geq -\left(\sum_{i=1}^{n} x_i r_{it} - \mu_p\right), \\ y_t &\geq 0 \qquad \forall\, t = 1, \ldots, T, \end{aligned}$$

the objective becomes:

$$\text{minimize} \quad \frac{1}{T}\sum_{t=1}^{T} y_t.$$

Every term is now linear in (x_i, y_t) - a standard LP.

Reference: Konno, H., & Yamazaki, H. (1991). *Mean-Absolute Deviation Portfolio*

Optimization Model and Its Applications to Tokyo Stock Market. Management Science, 37(5), 519–531. **Read:** Section 2 (MAD model and equivalence to variance under normality) and Section 3 (LP reduction via auxiliary variables).

Consequences: MAD produces the same efficient portfolios as variance under normality, with the major practical advantage of LP-solvability - enabling real-time optimization over thousands of assets. The auxiliary-variable linearization technique reappears in the CVaR and CDaR optimization frameworks.

2.4 Skewness and Co-Skewness

Variance and MAD capture the width of the return distribution but say nothing about its shape. Two portfolios with identical means and variances can have dramatically different tail behaviors.

Motivating example. Consider two portfolios with the same expected return of 10% and variance $\sigma^2 \approx 4\%^2$:

- **Portfolio A:** Returns symmetric around 10% - sometimes +20%, sometimes 0%, roughly equally likely.
- **Portfolio B:** Returns cluster around 12–15% most of the time, but occasionall, rarely drop to −40%.

Portfolio B has a long left tail. Any rational investor prefers Portfolio A. Variance cannot distinguish them.

Definition 2.4 (Skewness)**.** The third standardized central moment of returns:

$$\mathcal{S} \;=\; E\left[\left(\frac{R-\mu}{\sigma}\right)^3\right] \;=\; \frac{E[(R-\mu)^3]}{\sigma^3}.$$

Cubing preserves the sign of deviations. Averaging over the cubed, normalized deviations:

- **Negative skewness:** Left tail is fatter - rare, large losses. Typical of most equity portfolios in practice.
- **Positive skewness:** Right tail is fatter - rare, large gains (lottery-like payoffs).
- **Zero skewness:** Symmetric distribution (e.g., normal).

Definition 2.5 (Co-Skewness)**.** The co-skewness of asset i with the market portfolio m is:

$$CS_i \;=\; E\big[(R_i-\mu_i)(R_m-\mu_m)^2\big]\,.$$

Co-skewness measures how asset i's returns behave when the market is in its tail. Negative co-skewness means asset i underperforms its own mean precisely when the

market is in an extreme state - amplifying investor losses at the worst possible time. Positive co-skewness means the asset outperforms during market extremes, providing a natural hedge.

The Three-Moment CAPM. Kraus and Litzenberger (1976) extended CAPM from two moments (mean, variance) to three (mean, variance, skewness). Their main result is the pricing equation:

$$E[R_i] - R_f \;=\; d_1\,\beta_i + d_2\,\gamma_i,$$

where:

$$\beta_i \;=\; \frac{\mathrm{Cov}(R_i,\,R_m)}{\mathrm{Var}(R_m)} \quad \text{(standard market beta - sensitivity to market)},$$

$$\gamma_i \;=\; \frac{E[(R_i-\mu_i)(R_m-\mu_m)^2]}{E[(R_m-\mu_m)^3]} \quad \text{(co-skewness beta)},$$

and d_2 is the **price of co-skewness risk**, estimated empirically to be negative.

γ_i measures asset i's contribution to the portfolio's overall skewness, normalized by the market's own skewness - exactly analogous to how β_i measures the contribution to variance normalized by market variance.

Reference: Kraus, A., & Litzenberger, R. (1976). *Skewness Preference and the Valuation of Risky Assets.* The Journal of Finance, 21, 1085–1094. **Read:** Introduction and Section I (three-moment CAPM derivation and motivation for co-skewness pricing).

Consequences: Two assets can share the same β but have very different co-skewness. The asset with more negative co-skewness must offer a higher expected return in equilibrium, because rational investors demand compensation for bearing risk that materializes precisely during market crashes. This formally justifies the observed "overpricing" of crash insurance (out-of-the-money put options) relative to standard CAPM predictions.

3 Quantile-Based / Tail Risk Measures

3.1 Value at Risk (VaR)

Definition 3.1 (Value at Risk)**.** Let L denote portfolio loss over a given time horizon and $\alpha \in (0,1)$ a confidence level. The Value at Risk at level α is:

$$\mathrm{VaR}_\alpha \;=\; \inf\{l \in \mathbb{R} : P(L > l) \le 1-\alpha\}\,.$$

Equivalently, VaR_α is the $(1-\alpha)$ quantile of the loss distribution - the threshold that is exceeded with probability at most $(1-\alpha)$.

A one-day 99% VaR of \$10M means: there is only a 1% chance of losing more than \$10M on any given day.

Delta-Normal Method. Duffie and Pan (1997) describe several VaR computation methods. The simplest, the delta-normal (variance–covariance) method, assumes portfolio returns are normally distributed with mean μ and standard deviation σ:

$$\text{VaR}_\alpha \;=\; -\mu + z_\alpha\,\sigma,$$

where $z_\alpha = \Phi^{-1}(\alpha)$ is the α-quantile of the standard normal, i.e. $P(Z \le z_\alpha) = \alpha$ for $Z \sim \mathcal{N}(0,1)$.

Common values: $z_{0.95} = 1.645,\;\; z_{0.99} = 2.326$.

Square Root of Time Rule. Under i.i.d. normally distributed daily returns, VaR scales with the square root of the time horizon:

$$\text{VaR}_{n\text{-day}} \;=\; \text{VaR}_{1\text{-day}} \times \sqrt{n}.$$

Reference: Duffie, D., & Pan, J. (1997). *An Overview of Value at Risk*. The Journal of Derivatives, 4(3), 7–49. **Read:** Section 1 (definition and motivation) and Section 2 (delta-normal computation method and its normality assumption).

Consequences: VaR's strength is communicability - a single, intuitively interpretable loss number that became the global regulatory standard under Basel II. However, VaR has two fundamental weaknesses:

1. **Blindness to tail severity:** VaR identifies where the tail begins but says nothing about losses inside the tail. Two portfolios with identical 99% VaR may have catastrophically different loss profiles beyond that threshold.
2. **Non-subadditivity:** VaR can violate $\rho(X+Y) \le \rho(X) + \rho(Y)$. It can make diversification appear to increase measured risk - a fundamental incoherence formally demonstrated by Artzner et al. (1999).

3.2 Conditional Value at Risk (CVaR)

CVaR directly fixes both weaknesses of VaR.
Definition 3.2 (Conditional Value at Risk)**.**

$$\text{CVaR}_\alpha \;=\; E[L \mid L > \text{VaR}_\alpha]\,.$$

The expected loss given that the loss has already exceeded VaR_α - the average loss in the worst $(1-\alpha)\%$ of outcomes.

CVaR is also known as Expected Shortfall (ES), Mean Excess Loss, or Tail Value at Risk.

The Rockafellar–Uryasev Auxiliary Function. The challenge of minimizing CVaR is that VaR - which defines the conditioning threshold - is a non-smooth, non-convex function of portfolio weights, making direct optimization intractable. Rockafellar and Uryasev (2000) resolved this by introducing:

Definition 3.3 (F_α Auxiliary Function)**.**

$$F_\alpha(x,\,\zeta) \;=\; \zeta + \frac{1}{1-\alpha}\,E\big[\max\big(L(x)-\zeta,\;0\big)\big]\,,$$

where x is the vector of portfolio weights, $\zeta \in \mathbb{R}$ is a scalar threshold variable, $L(x)$ is the portfolio loss as a function of weights, and α is the confidence level.

The term $\frac{1}{1-\alpha}E[\max(L(x)-\zeta,0)]$ is the average excess loss beyond ζ, scaled up by $\frac{1}{1-\alpha}$ to account for the rarity of the tail.

Theorem 3.1 (Rockafellar–Uryasev, 2000)**.** $F_\alpha(x,\zeta)$ is jointly convex in (x,ζ). Minimizing $F_\alpha(x,\zeta)$ over both x and ζ simultaneously yields:

1. The optimal portfolio weights x^* that minimize CVaR.
2. $\zeta^* = \text{VaR}_\alpha$ as a by-product.
3. The minimum CVaR value directly.

VaR never needs to be computed as a separate step - it emerges automatically. Since F_α is jointly convex, any gradient-based optimizer is guaranteed to find the global minimum, not merely a local one.

LP Reformulation. Using T historical or simulated scenarios, introducing auxiliary variables $u_t \geq 0$:

$$F_\alpha(x,\zeta) \;\approx\; \zeta + \frac{1}{T(1-\alpha)}\sum_{t=1}^{T} u_t,$$

subject to:

$$u_t \geq L_t(x) - \zeta, \qquad u_t \geq 0, \qquad \forall\, t = 1,\ldots,T.$$

Every term is now linear in (x,ζ,u_t) - a standard LP, analogous to the MAD linearization in Section 2.3.

Why Minimize CVaR Instead of VaR?

VaR is not a convex function of portfolio weights - it may have local minima that are not global, making gradient-based optimization unreliable. CVaR is convex (through F_α), guaranteeing that any local minimum found is the global minimum.

Advantages of CVaR:

1. **Not blind to tail severity:** CVaR is defined as the average over the tail - tail content is the entire point.
2. **Subadditivity:** $\text{CVaR}_\alpha(X+Y) \leq \text{CVaR}_\alpha(X) + \text{CVaR}_\alpha(Y)$, which follows from the joint convexity of F_α.

CVaR satisfies all four coherence axioms of Artzner et al. (1999), which is precisely why Basel III / FRTB (2019) replaced 99% VaR with 97.5% Expected Shortfall as the primary regulatory measure.

Under Normality. For normally distributed losses with mean μ and standard deviation σ:

$$\text{CVaR}_\alpha \;=\; -\mu + \sigma \cdot \frac{\phi(z_\alpha)}{1-\alpha},$$

where ϕ is the standard normal PDF evaluated at the α-quantile z_α. CVaR is always strictly greater than VaR at the same α - it is the average of everything beyond the VaR threshold.

Reference: Rockafellar, R.T., & Uryasev, S. (2000). *Optimization of Conditional Value-at-Risk*. Journal of Risk, 2(3), 21–41. **Read:** Section 2 (the F_α construction and the joint-convexity theorem that makes optimization tractable).

Limitation: CVaR depends on the average of tail observations - the rarest, most extreme events. With limited data (e.g., 500 days at 99% confidence yields only 5 tail observations), a single outlier can swing the CVaR estimate dramatically. This estimation instability in the tail directly motivates the EVT approach in Section 6.

4 Drawdown-Based Measures

4.1 Maximum Drawdown and Conditional Drawdown at Risk (CDaR)

All measures in Sections 2 and 3 are single-period: they characterize the distribution of returns or losses at a point in time, ignoring the path through which losses accumulate. Drawdown measures capture this path-dependent experience, which drives the real-world phenomena of investor redemptions, margin calls, and stop-loss triggers.

Definition 4.1 (Drawdown)**.** The portfolio drawdown at time t is:

$$DD(t) \;=\; \frac{\max_{s \leq t} W(s) - W(t)}{\max_{s \leq t} W(s)},$$

where $W(t)$ is portfolio value at time t. It measures how far the portfolio currently

sits below its own historical peak. $DD(t) \geq 0$ always; $DD(t) = 0$ when the portfolio is at an all-time high.

Definition 4.2 (Maximum Drawdown (MDD))**.**

$$MDD \;=\; \max_{t\in[0,T]} DD(t).$$

The single worst drawdown value over the entire period.

Definition 4.3 (Average Drawdown (ADD))**.**

$$ADD \;=\; \frac{1}{T}\int_0^T DD(t)\,dt.$$

The time-averaged drawdown over the period.

MDD is highly sensitive to outliers - a single catastrophic episode dominates the statistic, analogous to VaR being a single quantile that ignores tail content. ADD smoothes out this sensitivity but can mask severe episodes by averaging them with mild periods. CDaR provides the principled middle ground.

The Underwater Curve. Chekhlov, Uryasev, and Zabarankin (2005) introduced the underwater curve - the time series $\{DD(t) : t \in [0, T]\}$, i.e., the drawdown plotted through time. When the curve is at 0%, the portfolio is at an all-time high; any positive value indicates the portfolio is underwater.

Definition 4.4 (Conditional Drawdown at Risk (CDaR))**.**

$$\mathrm{CDaR}_\alpha \;=\; \frac{1}{(1-\alpha)\,T}\int_{\text{worst }(1-\alpha)\text{ fraction}} DD(t)\,dt.$$

The average of the worst $(1 - \alpha) \times 100\%$ of drawdown observations along the path.

The parameter α unifies MDD and ADD as limiting cases:

- As $\alpha \to 0$: $\mathrm{CDaR}_\alpha \;\to\; ADD$ (average over all drawdown observations).
- As $\alpha \to 1$: $\mathrm{CDaR}_\alpha \;\to\; MDD$ (average over the single worst observation).

CDaR thus generalizes MDD the same way CVaR generalizes VaR.

Optimization Formulation. Chekhlov et al. observed that the CDaR formula is structurally identical to CVaR, simply applied to the drawdown path. The Rockafellar–Uryasev auxiliary function extends directly:

$$F_\alpha(x,\,\zeta) \;=\; \zeta + \frac{1}{(1-\alpha)\,T}\sum_{t=1}^{T} \max\big(DD_t(x) - \zeta,\; 0\big),$$

where x is the vector of portfolio weights, ζ is the threshold (equaling Drawdown-

at-Risk at the optimum), and $DD_t(x)$ is the drawdown at time t for portfolio x. Introducing auxiliary variables $u_t \geq \max(DD_t(x) - \zeta, 0)$ again yields an LP.

Reference: Chekhlov, A., Uryasev, S., & Zabarankin, M. (2005). *Drawdown Measure in Portfolio Optimization.* International Journal of Theoretical and Applied Finance, 8(1), 13–58. **Read:** Section 2 (underwater curve definition, CDaR construction, and proof of limiting cases).

Consequences for Portfolio Management:

1. **Redemption risk is path-dependent.** Investors redeem because they have watched losses mount for months, not because the return distribution has a fat tail. CDaR directly models the sustained-loss experience that triggers redemptions.

2. **Stop-loss rules are path-dependent.** Most institutional risk management policies specify "liquidate if drawdown exceeds $X\%$" - a constraint on MDD or CDaR, not on VaR. Single-period measures are structurally incapable of capturing this rule.

3. **Leverage and margin calls.** If a portfolio is leveraged, a sustained drawdown erodes equity, forcing deleveraging at exactly the wrong moment. CDaR captures the severity and duration of these periods; VaR and CVaR are entirely blind to them.

5 Time-Varying Volatility

All measures in Sections 2–4 implicitly treat risk as constant through time, using a single fixed σ estimated from historical data. Real financial return volatility exhibits **clustering**: turbulent periods are followed by turbulent periods, and calm periods by calm periods. ARCH and GARCH models make σ_t time-varying.

5.1 ARCH

Engle (1982) proposed the **Autoregressive Conditional Heteroscedasticity** (ARCH) model. Let r_t be the portfolio return at time t:

$$r_t \;=\; \mu + \varepsilon_t, \qquad \varepsilon_t \;=\; \sigma_t \, z_t,$$

where μ is the mean return, ε_t is the shock at time t, σ_t is the conditional volatility at time t, and $z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$.

Definition 5.1 (ARCH(q) Model)**.**

$$\sigma_t^2 \;=\; \omega + \sum_{i=1}^{q} \alpha_i \, \varepsilon_{t-i}^2$$

with $\omega > 0$, $\alpha_i \geq 0$, and $\sum_{i=1}^{q} \alpha_i < 1$ for stability.

Parameters:

- $\omega > 0$: baseline (long-run) variance; volatility never collapses to zero.
- ε_{t-i}^2: squared shock from i periods ago - a large past move in either direction increases today's estimated variance.
- α_i: weight assigned to the shock from i periods ago; recent shocks are typically given more weight.

Reference: Engle, R.F. (1982). *Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation*. Econometrica, 50(4), 987–1007. **Read:** The section specifying the ARCH(q) model and the conditional variance equation.

Limitation: To capture volatility clustering that lasts weeks or months, ARCH requires many lags q - a statistically wasteful and numerically unstable parameterization. Bollerslev (1986) covers this with GARCH.

5.2 GARCH

Bollerslev (1986) generalized ARCH by also allowing past variance (not just past squared shocks) to feed into current variance.

Definition 5.2 (GARCH(p,q) Model)**.**

$$\sigma_t^2 \;=\; \omega + \sum_{i=1}^{q} \alpha_i \, \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \, \sigma_{t-j}^2.$$

Definition 5.3 (GARCH(1,1) - The Workhorse)**.**

$$\sigma_t^2 \;=\; \omega + \alpha \, \varepsilon_{t-1}^2 + \beta \, \sigma_{t-1}^2.$$

Parameters:

- $\alpha \geq 0$: **reaction coefficient** - how strongly today's volatility reacts to yesterday's shock. High α means volatility jumps sharply after large moves.
- $\beta \geq 0$: **persistence coefficient** - how much today's volatility remembers yesterday's level. High β means elevated volatility persists for a long time.

- Stability condition: $\alpha + \beta < 1$.

Why GARCH(1,1) Dominates ARCH(q)?
By recursive substitution, GARCH(1,1) is equivalent to an infinite-order ARCH with geometrically decaying weights:

$$\sigma_t^2 \;=\; \frac{\omega}{1-\beta} + \alpha \sum_{i=1}^{\infty} \beta^{i-1} \varepsilon_{t-i}^2.$$

GARCH(1,1) achieves what ARCH(∞) would require with just **three parameters** (ω, α, β).

Long-Run (Unconditional) Variance. At steady state, setting $E[\sigma_t^2] = E[\sigma_{t-1}^2] = \bar{\sigma}^2$ and $E[\varepsilon_{t-1}^2] = \bar{\sigma}^2$:

$$\bar{\sigma}^2 \;=\; \omega + \alpha\bar{\sigma}^2 + \beta\bar{\sigma}^2 \qquad \Longrightarrow \qquad \boxed{\bar{\sigma}^2 \;=\; \frac{\omega}{1-\alpha-\beta}.}$$

After a shock drives volatility up, GARCH predicts it will revert toward $\bar{\sigma}^2$ at a speed governed by how close $(\alpha+\beta)$ is to 1. Typical equity index estimates yield $\alpha+\beta \approx 0.99$, implying highly persistent volatility.

Connection to VaR, CVaR, and CDaR.

- **Dynamic VaR:**

$$\mathrm{VaR}_{\alpha,t} \;=\; \mu + z_\alpha \cdot \hat{\sigma}_t, \qquad P(Z \le z_\alpha) = \alpha, \quad Z \sim \mathcal{N}(0,1),$$

  where $\hat{\sigma}_t$ is today's GARCH volatility estimate. During calm markets VaR shrinks; during turbulent periods it expands - far more responsive than static VaR.

- **Dynamic CVaR:** Same idea - replace fixed σ with $\hat{\sigma}_t$ in the CVaR formula.

- **CDaR:** GARCH identifies periods of elevated volatility, which are precisely when sustained drawdowns and margin calls are most likely.

Reference: Bollerslev, T. (1986). *Generalized Autoregressive Conditional Heteroskedasticity.* Journal of Econometrics, 31(3), 307–327. **Read:** The GARCH(p,q) specification and the argument showing it nests ARCH as a special case.

Limitations:

1. **Smooth volatility evolution.** GARCH updates σ_t^2 deterministically and gradually. It has no mechanism for sudden, discontinuous jumps - flash crashes, surprise central bank announcements - that shift the volatility level instantaneously

in a way unrelated to the smooth recursion.

2. **Long memory.** GARCH predicts volatility autocorrelation decays at exponential rate $(\alpha + \beta)^k$. Real financial data exhibits much slower, power-law decay - volatility has memory stretching years into the past - which GARCH systematically underestimates.

6 Extreme Value Theory for Tail Risk

In all measures covered so far, we implicitly assume we know (or can reliably estimate) the full distribution of returns. EVT challenges this assumption specifically for the tail.

The fundamental problem. Extreme losses are rare by definition. With 5 years of daily data ($\approx$ 1250 observations), a 99% confidence level leaves only 12–13 observations in the tail. Estimating tail behavior from 12 data points is statistically unreliable. For once in a decade events, you may have zero historical examples.

EVT's insight: You do not need to know the full return distribution to model the tail. The tail has a universal shape regardless of the underlying distribution, a mathematical theorem, not an assumption.

Theorem 6.1 (Central Limit Theorem vs Extreme Value Theorem)**.** .
CLT: The average of many i.i.d. random variables converges to a normal distribution, regardless of the original distribution.

EVT (Extreme Value Theorem): The maxima (or tail exceedances) of many i.i.d. random variables converge to a specific family of distributions, regardless of the original distribution.

6.1 Block Maxima: Generalized Extreme Value (GEV) Distribution

Divide data into blocks (e.g., monthly). Take the maximum loss in each block.
Definition 6.1 (GEV Distribution)**.** Block maxima follow the Generalized Extreme Value distribution:

$$G(x) \;=\; \exp\left\{ -\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi} \right\},$$

where μ is the location parameter (center), $\sigma > 0$ is the scale parameter (spread), and ξ is the **shape parameter** (tail heaviness).

ξ	Distribution Type	Tail Behavior
$\xi > 0$	Fréchet	Heavy/fat tail (power-law decay)
$\xi = 0$	Gumbel	Light tail (exponential decay)
$\xi < 0$	Weibull	Bounded tail (finite maximum)

For financial returns, $\xi > 0$ almost universally - confirming fat tails.

Problem. Block maxima waste information: if a month contains three extreme losses, only the single worst is used; the other two are discarded.

6.2 Peaks Over Threshold (POT): Generalized Pareto Distribution (GPD)

McNeil and Frey (2000) use the **Peaks Over Threshold (POT)** approach. Instead of block maxima, use all observations exceeding a high threshold u. For each exceedance, measure the **excess**:

$$y \;=\; x - u \qquad \text{given } x > u.$$

Theorem 6.2 (Pickands–Balkema–de Haan)**.** As threshold u becomes large, the distribution of excesses y converges to a **Generalized Pareto Distribution (GPD)**:

$$H_{\xi,\beta}(y) \;=\; \begin{cases} 1 - \left(1 + \dfrac{\xi\, y}{\beta}\right)^{-1/\xi} & \text{if } \xi \neq 0, \\ 1 - e^{-y/\beta} & \text{if } \xi = 0, \end{cases} \qquad \beta > 0.$$

This is the CDF: $P(\hat{z}_t < u + y \mid \hat{z}_t > u) \approx H_{\xi,\beta}(y)$.

The corresponding **survival function** (probability of exceeding $u + y$ given we are already above u) is:

$$P(\hat{z}_t > u + y \mid \hat{z}_t > u) \;\approx\; 1 - H_{\xi,\beta}(y) \;=\; \left(1 + \frac{\xi\, y}{\beta}\right)^{-1/\xi}.$$

The McNeil–Frey Two-Step Procedure. Neither GARCH nor EVT alone is sufficient: GARCH captures time-varying volatility but assumes normal innovations; EVT models the tail accurately but requires i.i.d. data, which raw financial returns are not (due to volatility clustering). McNeil and Frey (2000) combine them:

Step 1 - GARCH Filtering
Fit a GARCH(1,1) model to the return series $r_t = \mu + \sigma_t z_t$. Extract standardized

residuals:

$$\hat{z}_t \;=\; \frac{r_t - \hat{\mu}}{\hat{\sigma}_t}.$$

If GARCH fits well, $\hat{z}_t$ are approximately i.i.d. - the volatility clustering has been absorbed into $\hat{\sigma}_t$. EVT requires i.i.d. data to be valid; GARCH makes the data suitable for EVT.

Step 2 - EVT on Residual Tail

Fit a GPD to the tail excesses of $\hat{z}_t$ above threshold u. For a given standardized loss level z:

$$P(\hat{z}_t > z) \;=\; \frac{n_u}{n}\left(1 + \frac{\xi(z-u)}{\beta}\right)^{-1/\xi},$$

where n_u is the number of observations above u and n is the total number of observations.

Dynamic VaR from GPD

Inverting to obtain VaR_α in standardized residual space:

$$\widehat{\mathrm{VaR}}_\alpha^{z} \;=\; u + \frac{\hat{\beta}}{\hat{\xi}}\left[\left(\frac{n_u}{n(1-\alpha)}\right)^{\hat{\xi}} - 1\right].$$

Scaling back to actual return space:

$$\widehat{\mathrm{VaR}}_{\alpha,t} \;=\; \hat{\mu} + \hat{\sigma}_t \cdot \widehat{\mathrm{VaR}}_\alpha^{z}.$$

Dynamic CVaR from GPD

Using the mean-excess property of the GPD (conditioning on exceeding any level v yields another GPD with the same ξ and rescaled β, giving a linear mean excess function):

$$\widehat{\mathrm{CVaR}}_{\alpha,t} \;=\; \hat{\mu} + \hat{\sigma}_t\left(\frac{\widehat{\mathrm{VaR}}_\alpha^{z}}{1-\hat{\xi}} + \frac{\hat{\beta} - \hat{\xi}\,u}{1-\hat{\xi}}\right).$$

How to Choose the Threshold?

For each candidate threshold u, plot the **Mean Excess Plot**:

$$e(u) \;=\; E[X - u \mid X > u].$$

For a GPD, $e(u)$ is linear in u within the valid GPD range. Choose u at the point where this linearity begins in the empirical plot.

Reference: McNeil, A.J., & Frey, R. (2000). *Estimation of Tail-Related Risk Measures for Heteroscedastic Financial Time Series: An Extreme Value Approach.* Journal

of Empirical Finance, 7(3–4), 271–300. **Read:** Section 1 (why normality fails for tail estimation) and Section 2 (the two-step GARCH–EVT procedure).

Limitations:

1. **Threshold sensitivity:** Results depend meaningfully on the choice of u, introducing an element of subjectivity.
2. **Non-stationarity:** EVT assumes the tail distribution is stable over time. The GARCH step handles time-varying scale, but the shape parameter ξ is assumed constant - questionable across structurally different market regimes.

7 Consequences and Comparison of Measures

7.1 A Unified Narrative

The measures surveyed in this document form a coherent progression, each addressing a specific gap left by its predecessor:

1. **Markowitz (1952)** operationalized risk as variance - symmetric, tractable, foundational. But it penalizes upside equally with downside.

2. **Semi-variance (Markowitz, 1959)** fixed the symmetry problem conceptually but was computationally intractable.

3. **MAD (Konno & Yamazaki, 1991)** restored computational tractability via LP reformulation while remaining equivalent to variance under normality.

4. **Skewness & Co-skewness (Kraus & Litzenberger, 1976)** captured tail asymmetry entirely invisible to second-moment measures and showed it is a separately priced risk factor in equilibrium.

5. **VaR (Duffie & Pan, 1997)** moved from distributional summaries to a direct answer to "what can I lose?" - intuitive and regulatory-standard, but blind to tail severity and not subadditive.

6. **CVaR (Rockafellar & Uryasev, 2000)** fixed both VaR flaws - tail-complete and subadditive - replacing VaR as the Basel III regulatory standard.

7. **CDaR (Chekhlov et al., 2005)** moved from single-period to path-dependent measures, capturing drawdown dynamics that drive margin calls and redemptions that CVaR cannot see.

8. **GARCH (Bollerslev, 1986)** made all of the above time-varying: risk is not constant, and treating it as such systematically mis-measures it in both calm

and turbulent periods.

9. **EVT (McNeil & Frey, 2000)** addressed the final gap - reliable tail estimation without the normality assumption, by fitting the theoretically guaranteed GPD to GARCH-filtered residuals.

7.2 Coherence Scorecard

The following table evaluates the primary risk measures against the four Artzner et al. coherence axioms:

Measure	**Monotone**	**Subadditive**	**Pos. Homog.**	**Trans. Inv.**
Variance σ^2	×	×	✓	×
Standard Deviation σ	×	✓	✓	×
Semi-Variance	×	×	✓	×
MAD	×	✓	✓	×
VaR	✓	×	✓	✓
CVaR / Expected Shortfall	✓	✓	✓	✓

✓= satisfies; ×= does not satisfy in general.
CDaR, GARCH, and EVT are path-dependent or distributional models, not risk measures in the Artzner sense, and are not scored here.

CVaR is the only measure in this survey satisfying all four axioms - a fully coherent risk measure. This directly explains its adoption as the Basel III regulatory standard.

7.3 Key Practical Consequences

Regulatory evolution. The shift from Basel II (99% VaR) to Basel III/FRTB (97.5% Expected Shortfall) is a direct consequence of the mathematical properties surveyed here. VaR's non-subadditivity allowed banks to accumulate tail risk that did not appear in regulatory capital calculations, contributing materially to the 2008 financial crisis. CVaR/ES fixed this by construction.

The normality problem. Variance, semi-variance, MAD, delta-normal VaR, and dynamic CVaR under GARCH-normal all assume tail decay at normal or near-normal speed. EVT reveals that real equity returns have power-law tails with $\xi \approx 0.2$–0.4, implying that 4-sigma events occur roughly annually rather than once per century as

normality predicts. Every normality-based risk model therefore systematically underestimates catastrophic risk.

Path-dependence matters. The entire single-period risk framework (VaR, CVaR, variance, MAD) is blind to the sequence in which losses occur. Yet institutional investors face redemptions, leveraged traders face margin calls, and risk policies impose stop-loss rules - all of which depend on the path of portfolio value, not just its distribution at a point in time. CDaR is the only measure in this survey that captures this path-dependence, which is why drawdown statistics are reported universally by practitioners even when not required by regulators.

Time-varying risk. A constant σ in a world of volatility clustering means overestimating risk during calm periods (under-investing) and underestimating it during turbulent periods (over-exposing). GARCH is the minimum viable model for making risk estimates temporally responsive to actual market conditions.

8 Conclusion

This document has surveyed the landscape of mathematical risk measurement in quantitative finance, tracing the progression from Markowitz's 1952 variance framework through to extreme value theory. Each measure addressed a specific inadequacy of its predecessor: semi-variance and MAD corrected variance's symmetry and computational limitations; skewness and co-skewness captured tail asymmetry invisible to second-moment measures; VaR converted abstract statistical properties into a communicable loss threshold; CVaR restored coherence and tail-completeness that VaR lacked; CDaR extended the framework to path-dependent risks that single-period measures cannot see; GARCH made risk estimation temporally adaptive to volatility clustering; and EVT provided statistically principled tail modeling without distributional assumptions.

The single most important lesson across all these developments is that **no single risk measure is universally sufficient.** Variance is tractable but symmetric. VaR is intuitive but incoherent. CVaR is coherent but path-blind. CDaR is path-aware but single-period in its distributional assumptions. GARCH is temporally adaptive but assumes smooth evolution. EVT is tail-precise but assumes tail stationarity. A rigorous risk management framework requires multiple measures used together - their weaknesses complementary, their collective strengths providing a rounded picture of the true risk landscape.

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